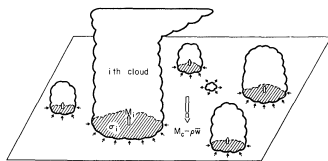


Step one of the theory: how a population of cumulus clouds changes the air around it.



$$\begin{aligned}\rho \partial_t \bar{s} &= D(\hat{s} - \bar{s} - L\hat{l}) + M_c \partial_z \bar{s} + LS_s \\ \rho \partial_t \bar{q} &= D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} + LS_q\end{aligned}$$

detrainment: subsidence: cools, moistens warms, dries large scale

Symbols

$s = c_p T + gz$ dry static energy, J kg^{-1}

q water-vapour mixing ratio, $\text{kg vapour} / \text{kg dry air}$

$\bar{s}, \bar{q}$ large-scale means, i.e. the environment

$M_c = \sum_i \rho \sigma_i w_i$ cloud mass flux, $\text{kg m}^{-2} \text{s}^{-1}$

σ_i, w_i area fraction, vertical velocity of cloud i

D mass detrained per unit volume and time

$\hat{\cdot}, \hat{\cdot}^*$ in the detrained air; at saturation

$\hat{l}$ liquid water in the detrained air

L, ρ latent heat of vaporization; air density

LS large-scale advection (+ radiation for s)

Clouds act on the environment only through M_c , D and $\hat{l}$

Each cloud type is an entraining plume

2

The clouds are sorted into types by how fast they mix in outside air, the rate λ . Given the environment, each type's mass flux, energy and top follow.

$$\partial_z \eta = \lambda \eta \quad \Rightarrow \quad \eta = e^{\lambda(z-z_B)}$$

$$\eta h_c = h_M + \lambda \int_{z_B}^z \eta \bar{h} dz'$$

$$h_c(z_D) = \hat{h}^* \approx \bar{h}^* \quad (\text{top})$$

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

Symbols

λ fractional entrainment rate, km^{-1} : the type label. Large

λ : small, diluted cloud with a low top (Fig. 8)

$\eta(z, \lambda)$ mass flux of type λ , normalized to 1 at z_B

$h = s + Lq$ moist static energy: conserved also when vapour condenses

h_c, h_M h inside the cloud; in the subcloud mixed layer

$\bar{h}^*$ h of the environment if it were saturated

z_B top of the mixed layer, where the clouds take their air

z_D cloud top, where buoyancy vanishes

$\mathcal{M}_B(\lambda)$ mass flux of type λ at z_B

$\lambda_D(z)$ type whose top is at height z

The environment fixes each plume: only the spectrum $\mathcal{M}_B(\lambda)$ is unknown

Cloud work function and quasi-equilibrium

3

$A(\lambda)$ is the kinetic energy that buoyancy gives clouds of type λ , per unit mass flux. The large scale builds it up; the clouds use it up.



$$A(\lambda) = \int_{z_B}^{z_D(\lambda)} \eta \underbrace{\frac{g(s_{vc} - \bar{s}_v)}{c_p \bar{T}}}_{\text{buoyancy}} dz$$

$$\frac{dA}{dt} = \int K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) \simeq 0$$

$\simeq 0$: quasi-equilibrium, clouds adjust much faster than F changes.

Symbols

$s_v = s + c_p \bar{T}(0.608 q - l)$, virtual static energy: vapour makes air lighter, liquid water l heavier

$s_{vc}, \bar{s}_v$ in the cloud; in the environment

$g, \bar{T}$ gravity; environment temperature

$A > 0$ type λ can grow. For $\lambda = 0$, A is close to CAPE

$K(\lambda, \lambda')$ change of $A(\lambda)$ per unit $\mathcal{M}_B(\lambda')$; mostly negative

$F(\lambda)$ change of A by large-scale processes

Quasi-equilibrium turns $\mathcal{M}_B(\lambda)$ into the solution of an integral equation

Observations support quasi-equilibrium

4

Test with tropical data. If clouds did not respond, A would change as fast as the forcing (dashed line). The observed change of A stays near zero.

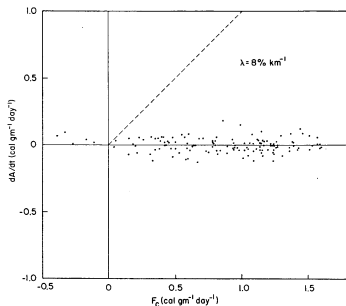


Fig. 13: observed dA/dt against F_C , for $\lambda = 8\% \text{ km}^{-1}$.

Two timescales

$$\tau_{\text{adj}} \sim 10^3 - 10^4 \text{ s}$$

15 min to 3 h

$$\tau_{\text{LS}} \gtrsim 10^5 \text{ s}$$

a day or more

Observed:

$$|dA/dt| \ll |F_C|$$

Symbols

F_C change of A that large-scale processes in the cloud layer would produce alone

dA/dt observed change of A

τ_{adj} time for the clouds to remove A

τ_{LS} time over which the large-scale forcing changes

cal g^{-1} energy per unit mass; 1

$\text{cal g}^{-1} \approx 4.2 \text{ kJ kg}^{-1}$

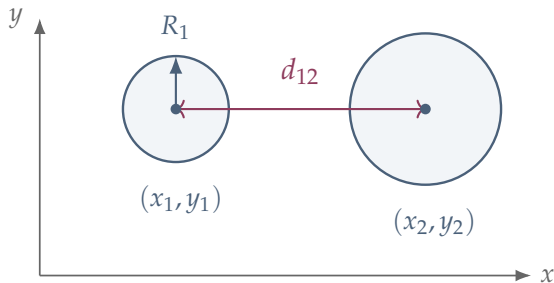
Clouds nearly cancel the forcing: A changes much less than F_C predicts

No 2D distribution of thermals

5

Our question: how are thermals arranged in a horizontal plane? The theory sorts clouds by type, not by position.

Spatial quantities (schematic)



Sought, not in the paper

R_i radius of thermal i at a fixed height

(x_i, y_i) position of its centre

d_{ij} distance between thermals i and j

Why the theory does not answer it

A spectrum over λ has no positions or distances.

$\mathcal{M}_B$ mixes cloud number, lifetime and single-cloud mass flux, never separated.

The environment is assumed horizontally uniform.

A mass-flux spectrum leaves the horizontal thermal field unspecified